

BEATRIZ SOUZA

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EDUCATION

APR 2023-now	University of Stuttgart , Germany PhD in Computer Science Advisor: Michael Pradel Research Area: AI4Code, Execution, Program Analysis
OCT 2021-FEB 2023	Federal University of Pernambuco , Brazil Master's in Computer Science Advisor: Marcelo d'Amorim Thesis: Learning to Detect Text-Code Inconsistencies with Weak and Manual Supervision
MAY 2017-JUNE 2021	Federal University of Campina Grande , Brazil Bachelor's in Computer Science Advisor: Rohit Gheyi Thesis: Most Higher Order Mutants are Useless for Method Level Operators
MAY 2013-SEPT 2016	Federal Institute of Education, Science and Technology of Paraíba , Brazil High School with a Technical Degree in Computer Science Advisor: Gustavo Vieira

WORK EXPERIENCE

2024 (APR-JULY)	Microsoft Research , Redmond, WA, USA - RESEARCH INTERN Mentor and Collaborator: Suman Nath and Chang Lou Project: <i>runtime verification</i> of distributed systems.
2022 (JULY-DEC)	Software Lab , Stuttgart, Germany - RESEARCH INTERN Advisor: Michael Pradel Project: <i>LExecutor</i> , a learning-guided approach for executing arbitrary Python code snippets.
MAY 2020-FEB 2021	Analytics Lab , Campina Grande, Brazil - SOFTWARE DEVELOPER Mentors: João Brunet and Nazareno Andrade Projects: <i>Tá de Pé?</i> and <i>Monitor Cidadão</i> , which are web applications aiming to identify risks in Brazilian government contracts.
NOV 2019-OCT 2020	UFCEG , Campina Grande, Brazil - RESEARCH ASSISTANT Advisor: Rohit Gheyi Project: I Worked on a sound and lightweight technique, based on theorem proving using Z3, to identify equivalent, duplicate, and subsumed mutations.
AUG 2018-JULY 2019	SPLAB , Campina Grande, Brazil - RESEARCH ASSISTANT Advisor: Patrícia Machado Project: I investigated the effectiveness of state-of-the-art tools that automatically generate test cases, such as Randoop and EvoSuite.
JULY 2016-SEPT 2016	Papagaio , João Pessoa, Brazil - WEB DEVELOPER INTERN Mentor: Gustavo Vieira Project: I participated in the development of an e-commerce platform for drug stores.
AUG 2015-JULY 2016	IFPB , Cajazeiras, Brazil - RESEARCH ASSISTANT Advisors: Gustavo Vieira and Wilza Moreira Project: I worked on <i>Segundo Mendel</i> , a mobile application to help high school students to learn genetic concepts, such as Mendel's Laws.

VOLUNTEER SERVICE

- Student volunteer of the ASE 2020 conference.

PUBLICATIONS

- ICSE'25 Treefix: Enabling Execution with a Tree of Prefixes.
Beatriz Souza and Michael Pradel. *International Conference on Software Engineering*
- FSE'25 ChangeGuard: Validating Code Changes via Pairwise Learning-Guided Execution.
Lars Gröninger, **Beatriz Souza** and Michael Pradel.
Symposium on the Foundations of Software Engineering
- FSE'23 LExecutor: Learning-Guided Execution.
Beatriz Souza and Michael Pradel. *Symposium on the Foundations of Software Engineering*
- IST'21 Identifying Method-Level Mutation Subsumption Relations using Z3.
Rohit Gheyi, Márcio Ribeiro, **Beatriz Souza**, Marcio Guimarães, Leo Fernandes, Marcelo d'Amorim, Vander Alves, Leopoldo Teixeira, Baldoino Fonseca. *Elsevier Information and Software Technology*
- SBES'20 A Large Scale Study On the Effectiveness of Manual and Automatic Unit Test Generation.
Beatriz Souza and Patrícia Machado. *Brazilian Symposium on Software Engineering*
- SBES'20 A Lightweight Technique to Identify Equivalent Mutants.
Beatriz Souza and Rohit Gheyi. *Brazilian Symposium on Software Engineering*
- ASE'20 Identifying Mutation Subsumption Relations.
Beatriz Souza. *International Conference on Automated Software Engineering*
- SPLASH'19 Is Mutation Score a Fair Metric?.
Beatriz Souza. *International Conference on Systems, Programming, Languages, and Applications: Software for Humanity*

AWARDS AND HONORS

- **Selected for the 12th Heidelberg Laureate Forum as a young researcher.**
- **ACM SIGSOFT Distinguished Paper Award at The Symposium on the Foundations of Software Engineering (FSE'23)** for the work *LExecutor: Learning-Guided Execution*.
- **Best paper at The Brazilian Symposium on Software Engineering (SBES'20)** for the work *A Large Scale Study On the Effectiveness of Manual and Automatic Unit Test Generation*.
- **First Place at The Undergraduate Research on Software Engineering Competition (SBES'20)** for the work *A Lightweight Technique to Identify Equivalent Mutants*.
- **Third Place at The ACM Student Research Competition (ASE'20)** for the work *Identifying Mutation Subsumption Relations*.
- **Third Place at The ACM Student Research Competition (SPLASH'19)** for the work *Is Mutation Score a Fair Metric?*.

TEACHING EXPERIENCE

APR 2025-SEPT 2025	Teaching assistant at SOLA, Stuttgart, Germany
OCT 2024-MAR 2025	Programming Paradigms - I created and graded exercise lists and addressed questions for a class of 80+ students.
APR 2023-SEPT 2023	
MAR 2018-AUG 2018	Teaching assistant at UFCG, Campina Grande, Brazil
	Mathematical Foundations for Computer Science - I helped professor by grading exercise lists and addressing questions for a class of 46 students.
JULY 2015-DEC 2015	Teaching assistant at IFPB, Cajazeiras, Brazil
	Competitive Programming - I collaborated in a course to prepare high school and undergraduate students for programming contests using C as programming language.

MENTORING

BACHELOR THESES

- Aleksis Vezenkov. Performance-Improving Refactorings for Python. 2025

MASTER THESES

- Lars Gröninger. Reasoning about Code Changes via Pairwise Learning-Guided Execution. 2024

Keep Running: Towards AI-Assisted Approaches for Enabling and Monitoring Program Execution

- Max Planck Institute for Software Systems. September, 2025.
- 12th Heidelberg Laureate Forum. September, 2025.

Learning-Guided Execution

- ELLIS Unit Stuttgart. July, 2025
- Microsoft Research Redmond. May, 2024